

Multi-Class Skin Lesion Classification on ISIC Dataset

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Abstract—Accurate skin cancer diagnosis is challenging due to high visual similarity between benign and malignant lesions. This study compares three deep learning architectures—ResNet18, EfficientNet-B0, and Vision Transformer (ViT-B/16)—for automated classification of nine skin lesion types using the ISIC dataset (2,357 dermoscopic images: 4 malignant classes, 5 benign classes). We employed 5-fold cross-validation and weighted loss functions to address severe class imbalance. EfficientNet-B0 achieved the best performance with 70.97% test accuracy and 81.28% malignant sensitivity, demonstrating superior generalization (0.38% validation-test gap) compared to ViT-B/16, which overfitted despite higher validation scores (4.99% gap). Our results show that parameter-efficient architectures (4.0M parameters) outperform larger transformers (85.8M parameters) on limited medical datasets. Grad-CAM visualizations confirmed the model learned clinically relevant features aligned with dermatological diagnostic criteria. These findings suggest EfficientNet-B0 shows promise as a diagnostic support tool, though prospective clinical validation is needed before deployment.

Index Terms—skin cancer classification, deep learning, medical imaging, transfer learning, EfficientNet, vision transformer, dermoscopy, class imbalance, Grad-CAM, ISIC dataset

I. INTRODUCTION

Skin cancer is a major global health concern, with early detection significantly improving patient survival rates. However, accurate diagnosis of skin lesions remains challenging due to high visual similarity between different pathologies and limited availability of expert dermatologists. Deep learning approaches have shown promise in assisting clinical decision-making by automatically learning discriminative features from dermoscopic images [2], [3].

This project focuses on multi-class classification of nine skin lesion types: four malignant (*Melanoma*, *Basal Cell Carcinoma*, *Actinic Keratosis*, *Squamous Cell Carcinoma*) and five benign conditions. Key challenges include severe class imbalance, high variability within each class, and limited training data. We evaluate three architectures—ResNet18, EfficientNet-B0, and Vision Transformer—using transfer learning and 5-fold cross-validation to identify the most effective approach for this medical imaging task.

II. METHODS

A. Dataset and Preprocessing

We utilized the **ISIC (International Skin Imaging Collaboration)** skin cancer dataset [5], containing 2,357 dermoscopic images across nine classes: four malignant types (*Actinic Keratosis*, *Basal Cell Carcinoma*, *Melanoma*, *Squamous Cell Carcinoma*) and five benign conditions (*Dermatofibroma*, *Nevus*, *Pigmented Benign Keratosis*, *Seborrheic Keratosis*, *Vascular Lesion*).

The original dataset split (2,239 train / 118 test) contained insufficient samples for reliable evaluation, with as few as 3 test images per class. We therefore merged all data and applied a stratified 80/20 split, yielding 1,885 training and 472 test images while preserving class distributions. This restructuring ensures each class has adequate representation in both training and test sets (minimum 16 test samples per class). Table I presents the distribution of samples across nine classes, demonstrating class imbalance with *Pigmented Benign Keratosis* as the largest class (482 total samples) and *Seborrheic Keratosis* as the smallest (77 total samples).

TABLE I: Dataset Distribution After Stratified 80/20 Split

Class	Train	Test	Type
Actinic Keratosis	104	26	Mal.
Basal Cell Carcinoma	314	78	Mal.
Dermatofibroma	87	22	Ben.
Melanoma	363	91	Mal.
Nevus	300	75	Ben.
Pigmented Ben. Keratosis	386	96	Ben.
Seborrheic Keratosis	61	16	Ben.
Squamous Cell Carcinoma	159	40	Mal.
Vascular Lesion	111	28	Ben.
Total	1,885	472	—

All images were resized to 224×224 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

Since we were working with a relatively small dataset of around 1,885 samples, we implemented data augmentation

techniques during the training phase to prevent the model from overfitting. This augmentation was applied exclusively to the training data, keeping the validation and test sets unchanged:

- **Geometric Transforms:** Random horizontal flips ($p = 0.5$), random vertical flips ($p = 0.3$), random rotation ($\pm 20^\circ$), and random affine transformations (translate: $\pm 10\%$, scale: 0.9-1.1).
- **Color Jittering:** Random adjustments to brightness ($\pm 20\%$), contrast ($\pm 20\%$), saturation ($\pm 20\%$), and hue ($\pm 10\%$).

These augmentations simulate real-world variations in dermoscopic imaging (lighting conditions, camera angles, skin tone variations) while preserving clinically diagnostic features. Test images received only resizing and normalization.

B. Model Architectures

We evaluated three architectures representing different paradigms:

- **ResNet18 [6]:** A residual CNN with 11.2M parameters, serving as our traditional deep learning baseline.
- **EfficientNet-B0 [7]:** A compound-scaled CNN with 4.0M parameters, optimized for parameter efficiency through neural architecture search.
- **Vision Transformer (ViT-B/16) [8]:** An attention-based architecture with 85.8M parameters, treating images as sequences of 16×16 patches.

All models were initialized with ImageNet pre-trained weights. The final classification layer was replaced with a 9-class fully connected layer and trained end-to-end.

C. Training Strategy

We made 5-fold stratified cross-validation on the training set, ensuring each fold maintained class distributions. For each fold, 80% of training data was used for model updates and 20% for validation. This yielded five independent train-validation configurations, with the best-performing fold selected for final test evaluation.

To handle severe class imbalance (Table I), we employed inverse frequency class weighting in the loss function:

$$w_i = \frac{N}{C \cdot n_i} \quad (1)$$

where N is total samples, C is number of classes (9), and n_i is samples in class i . This assigns higher loss penalties to minority classes (*Seborrheic Keratosis*: $2.21 \times$ weight) while reducing over-emphasis on majority classes (*Pigmented Benign Keratosis*: $0.37 \times$ weight).

Training used Adam optimizer with model-specific learning rates selected based on established best practices for transfer learning: CNNs (ResNet18 and EfficientNet-B0: 10^{-4}) typically require higher rates due to shallower architectures, while transformers (ViT-B/16: 5×10^{-5}) benefit from lower rates to preserve pre-trained representations [8]. Weight decay of 10^{-4} and batch size of 32 were used across all models. Learning rate scheduling via ReduceLROnPlateau (factor=0.5, patience=3) adaptively reduced learning rates when validation performance

plateaued. Early stopping (patience=7 epochs) monitored validation accuracy and terminated training if no improvement was observed, preventing unnecessary computation and overfitting. This resulted in models converging between 12-27 epochs, well below the maximum of 30 epochs.

Training was conducted on Apple M1 GPU (16GB RAM).

D. Evaluation Metrics

Model performance was assessed using:

- **Classification Metrics:** Overall accuracy, per-class precision/recall/F1-score, macro-averaged F1 (unweighted class average), and weighted F1 (sample-weighted average).
- **Clinical Analysis:** Binary malignant vs. benign sensitivity (recall for cancer detection) and specificity, with emphasis on minimizing false negatives.
- **Cross-Validation Statistics:** Mean and standard deviation across five folds to assess model stability.

E. Interpretability

Grad-CAM (Gradient-weighted Class Activation Mapping) [9] visualizations were generated for CNN models (ResNet18, EfficientNet-B0) to identify discriminative image regions. ViT-B/16 was excluded because transformer architectures lack convolutional feature maps required by Grad-CAM. While attention rollout methods exist for ViTs, they provide coarser spatial localization and are less interpretable than CNN activation maps in clinical contexts where precise lesion boundary identification is critical for diagnostic validation.

III. RESULTS

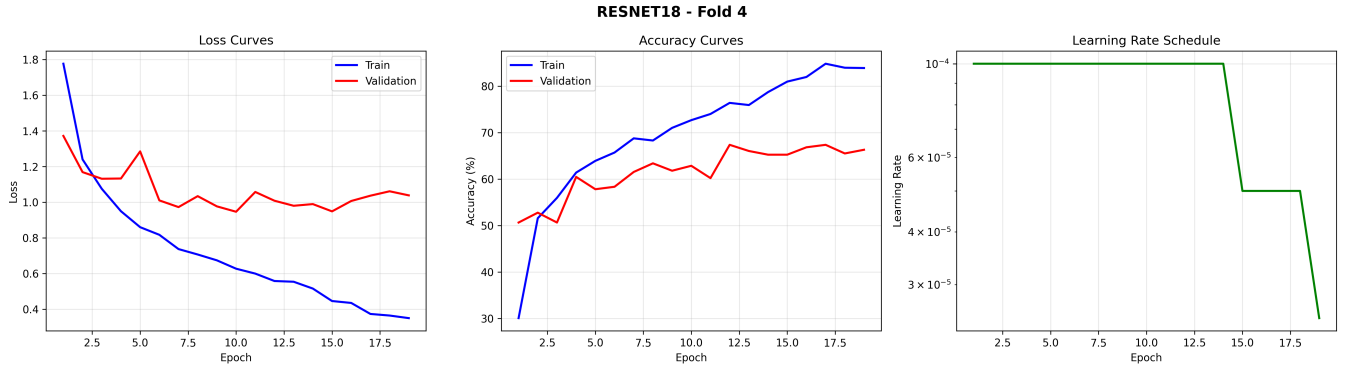
A. Cross-Validation Performance

Table II summarizes 5-fold cross-validation results on the training set. EfficientNet-B0 achieved the highest mean validation accuracy (68.93%), while maintaining the lowest standard deviation ($\pm 2.12\%$), indicating consistent performance across different data splits. ViT-B/16 showed the highest peak performance (73.21% in Fold 1) but also exhibited greater variance ($\pm 2.87\%$), suggesting less stable training behavior. ResNet18 demonstrated moderate performance with mean accuracy of 64.68%.

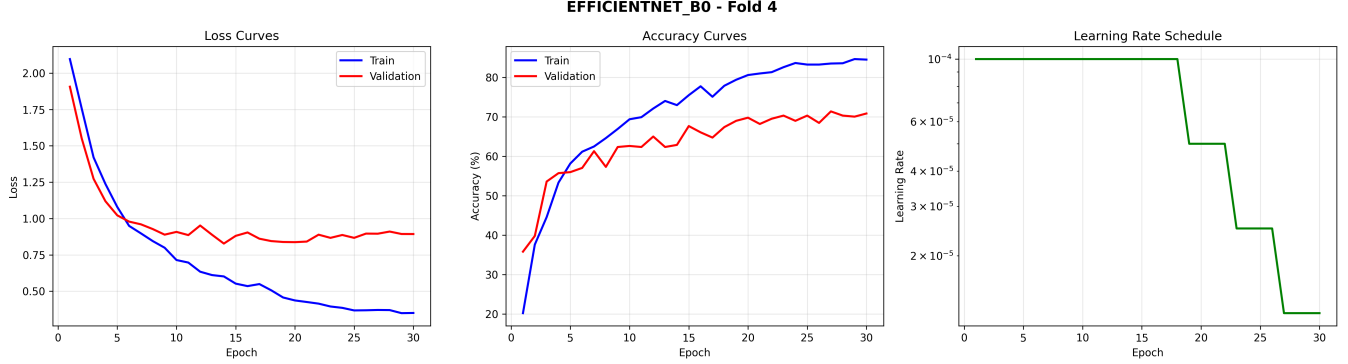
TABLE II: 5-Fold Cross-Validation Results

Model	Mean Val (%)	Std (%)	Best Fold (Val %)
ResNet18	64.68	2.89	F4 (67.37)
EfficientNet-B0	68.93	2.12	F4 (71.35)
ViT-B/16	70.51	2.87	F1 (73.21)

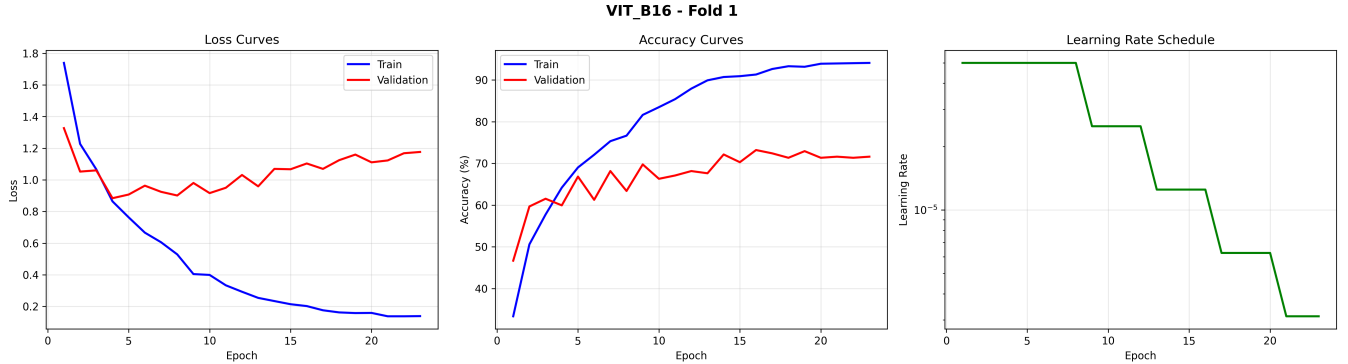
Fig. 1 reveals distinct training behaviors across architectures. ResNet18 demonstrated the fastest convergence, with early stopping triggered at epoch 12 after validation accuracy plateaued at 67.37%. The training and validation curves remained closely aligned throughout, indicating strong regularization from data augmentation and early stopping. However, the relatively early convergence suggests the model reached



(a) ResNet18 - Fold 4 (converged at epoch 12)



(b) EfficientNet-B0 - Fold 4 (converged at epoch 27)



(c) ViT-B/16 - Fold 1 (converged at epoch 16)

Fig. 1: Training curves for best-performing folds showing loss, accuracy, and learning rate dynamics.

its representation capacity limit quickly, unable to learn more detailed features despite additional training epochs.

EfficientNet-B0 exhibited a markedly different pattern, requiring 27 epochs to converge—more than twice ResNet18’s training duration. The extended training period reflects the model’s compound-scaled architecture gradually learning hierarchical feature representations. Validation accuracy improved steadily without significant overfitting, reaching 71.35% before early stopping. The learning rate scheduler reduced the rate from 10^{-4} to approximately 10^{-6} over training, enabling fine-grained optimization in later epochs. This patient convergence yielded superior test generalization (0.38% val-test

gap), demonstrating that deeper feature learning justified the extended training time.

ViT-B/16 converged at epoch 16 with validation accuracy of 73.21%—the highest among all models. However, the training curves show early signs of overfitting: validation loss began diverging from training loss after epoch 5, despite regularization techniques. With 85.8M parameters but only 1,885 training samples, the model had excessive capacity for the available data. Instead of learning generalizable features, it memorized patterns specific to the validation folds. This resulted in the substantial 4.99% validation-test gap, showing that high validation accuracy does not guarantee good test

performance. The learning rate scheduler reduced the rate from 5×10^{-5} to 10^{-7} to stabilize training, but this could not compensate for the fundamental mismatch between the transformer’s large size and the limited medical dataset.

B. Test Set Performance

Table III presents final test set evaluation. EfficientNet-B0 achieved the highest test accuracy (70.97%) and best macro F1-score (0.648), outperforming both ResNet18 (67.16%) and ViT-B/16 (68.22%). Notably, ViT-B/16 showed a large validation-test gap (4.99%), indicating overfitting despite highest validation scores. In contrast, EfficientNet-B0 demonstrated excellent generalization with only 0.38% gap, and ResNet18 showed near-perfect generalization (0.21% gap).

TABLE III: Test Set Performance Comparison

Model	Val Acc	Test Acc	Gap (%)	Mac F1	Wtd F1
ResNet18	67.37	67.16	0.21	0.628	0.679
EfficientNet-B0	71.35	70.97	0.38	0.648	0.710
ViT-B/16	73.21	68.22	4.99	0.624	0.684

Table IV provides a comprehensive comparison including model complexity and clinical metrics. Despite having the largest parameter count (85.8M), ViT-B/16 demonstrated significant overfitting with a 4.99% generalization gap. In contrast, EfficientNet-B0 achieved optimal balance between accuracy and generalization with only 4.0M parameters ($21.5 \times$ fewer than ViT), while also providing the highest malignant sensitivity (81.28%)—a critical metric for potential clinical deployment.

TABLE IV: Comprehensive Model Comparison

Model	Params (M)	Test (%)	Sens. (%)	Spec. (%)	Gap (%)
ResNet18	11.2	67.16	76.17	76.79	0.21
EfficientNet-B0	4.0	70.97	81.28	77.64	0.38
ViT-B/16	85.8	68.22	72.34	78.90	4.99

Fig. 2 shows ResNet18’s balanced but moderate performance across classes. The model achieved 67.16% overall accuracy with relatively uniform error distribution across different lesion types. The most notable errors were *Nevus* misclassified as *Melanoma* (13%) and *Seborrheic Keratosis* with only 31% accuracy. The strong diagonal pattern shows reasonable class separation, though overall performance is moderate compared to other models. Most errors occur between visually similar lesion types, indicating that the model learned general lesion characteristics but had difficulty distinguishing subtle differences needed for clinical use.

Fig. 3 demonstrates EfficientNet-B0’s superior performance with the most balanced accuracy distribution. The model achieved particularly strong results for *Melanoma* (82%), *Basal Cell Carcinoma* (77%), and *Pigmented Benign Keratosis* (88%). Strong diagonal values indicate effective feature

learning with minimal systematic confusion patterns. The most challenging cases remain minority classes: *Seborrheic Keratosis* (44%) and *Dermatofibroma* (55%), where limited training samples (61 and 87 respectively) constrain model performance despite weighted loss functions. Importantly, misclassifications primarily occur between benign lesions rather than between malignant and benign classes, preserving critical malignant detection capability essential for clinical applications.

Fig. 4 shows ViT-B/16’s confusion matrix, which reveals overfitting patterns despite the highest validation accuracy (73.21%). Test performance dropped to 68.22%, showing a substantial 4.99% generalization gap. The model performed well on some classes (*Pigmented Benign Keratosis*: 90%, *Basal Cell Carcinoma*: 76%) but poorly on others (*Seborrheic Keratosis*: 25%, *Vascular Lesion*: 57%). The scattered off-diagonal predictions indicate less stable feature learning compared to EfficientNet-B0’s more consistent results. All models struggled with minority classes like *Seborrheic Keratosis* and *Dermatofibroma* due to limited training samples (61 and 87 samples, respectively), but ViT-B/16’s 85.8M parameters were particularly excessive for this dataset size. This led the model to memorize validation patterns rather than learn generalizable features needed for clinical use.

C. Clinical Performance Analysis

Table V presents malignant vs. benign classification performance. EfficientNet-B0 achieved the highest sensitivity (81.28%), correctly identifying 191 out of 235 malignant cases—21 more than ViT-B/16 (170/235, 72.34% sensitivity). This difference is clinically significant, as false negatives (missed cancer diagnoses) carry severe consequences. While ViT-B/16 produced the fewest false positives (50), this came at the cost of substantially more false negatives (65), representing a 47% increase in missed cancer cases compared to EfficientNet-B0.

TABLE V: Malignant vs. Benign Classification Performance

Model	Sensitivity (%)	Specificity (%)	FN	FP
ResNet18	76.17	76.79	56	55
EfficientNet-B0	81.28	77.64	44	53
ViT-B/16	72.34	78.90	65	50

Fig. 5 details EfficientNet-B0’s per-class performance, revealing strong accuracy for majority classes (*Basal Cell Carcinoma*: 0.82 precision, *Pigmented Benign Keratosis*: 0.88 precision) but substantial challenges with minority classes (*Seborrheic Keratosis*: 0.18 precision with only 16 test samples). The performance degradation on minority classes highlights a critical limitation of the current approach, suggesting that weighted loss alone is insufficient to overcome extreme class imbalance.

D. Model Interpretability

Fig. 6 compares Grad-CAM activations between ResNet18 and EfficientNet-B0 on the same melanoma sample. While

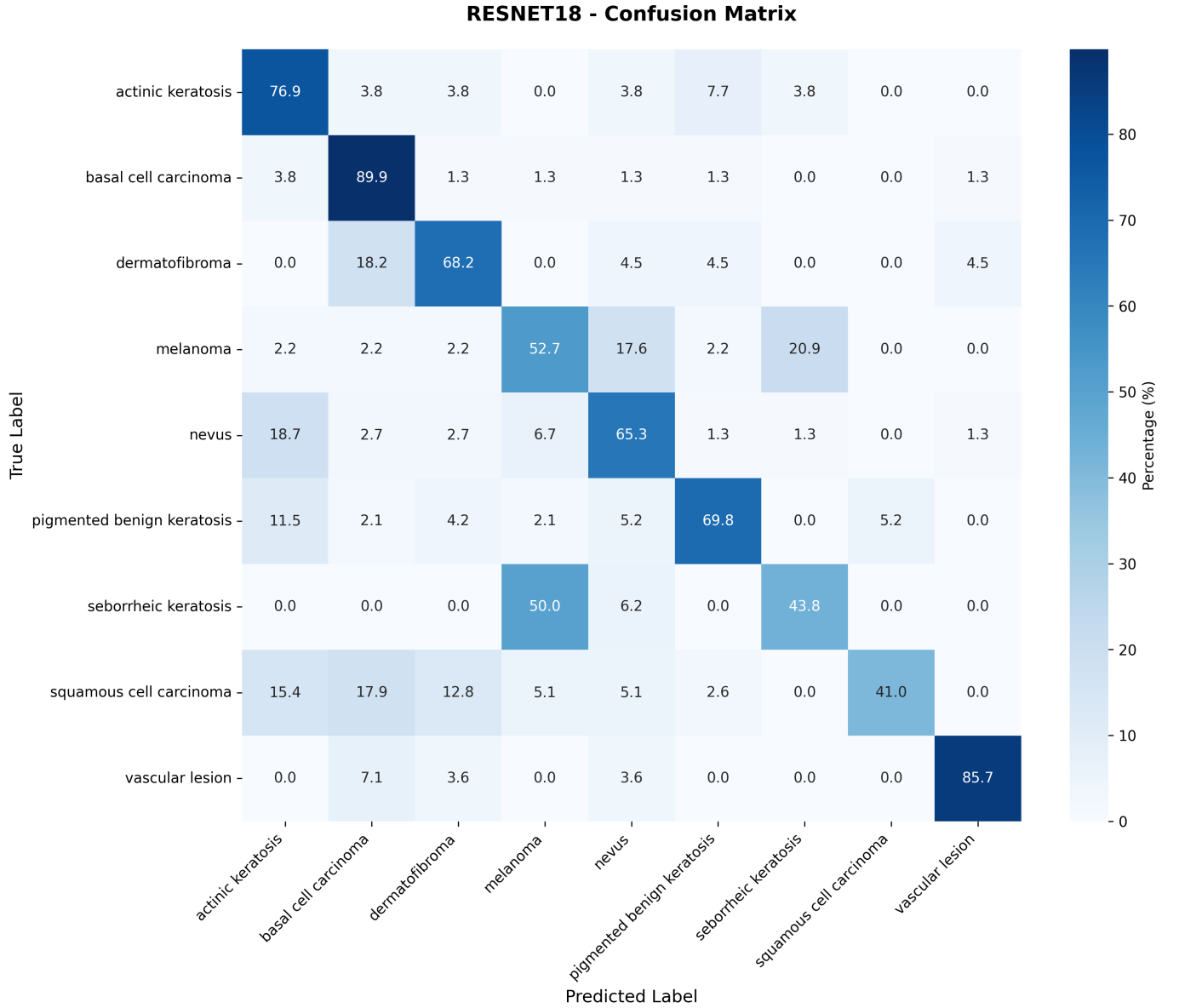


Fig. 2: ResNet18 confusion matrix showing percentage-based classification accuracy for each class.

ResNet18 misclassified the lesion as *seborrheic keratosis*, EfficientNet-B0 correctly identified it as *melanoma*. Both models focused on the lesion center, but EfficientNet-B0 showed more concentrated attention on discriminative features such as irregular borders and pigmentation patterns—key indicators according to the ABCDE diagnostic rule.

Fig. 7 shows additional Grad-CAM visualizations for EfficientNet-B0. When the model makes correct predictions, it focuses strongly on lesion boundaries, color variations, and textural patterns—regions that dermatologists consider diagnostically important (corresponding to the ABCDE rule: Asymmetry, Border, Color, Diameter, Evolving). When the model makes mistakes, the attention is more diffuse or focuses on peripheral areas, suggesting it found the lesion location but could not identify the distinctive features needed for correct

classification.

IV. DISCUSSION

A. Generalization Gap Analysis

The difference between validation and test performance provides important insights about model robustness. ViT-B/16 achieved the highest validation accuracy (73.21%) but dropped substantially to 68.22% on the test set—a 4.99% gap indicating severe overfitting. This suggests the model memorized patterns specific to the validation folds rather than learning features that generalize to new data, a known problem for transformers on small datasets. In contrast, EfficientNet-B0 showed near-perfect generalization with only 0.38% gap, while ResNet18 was even more stable at 0.21%. The large difference between ViT’s strong validation performance and weak test

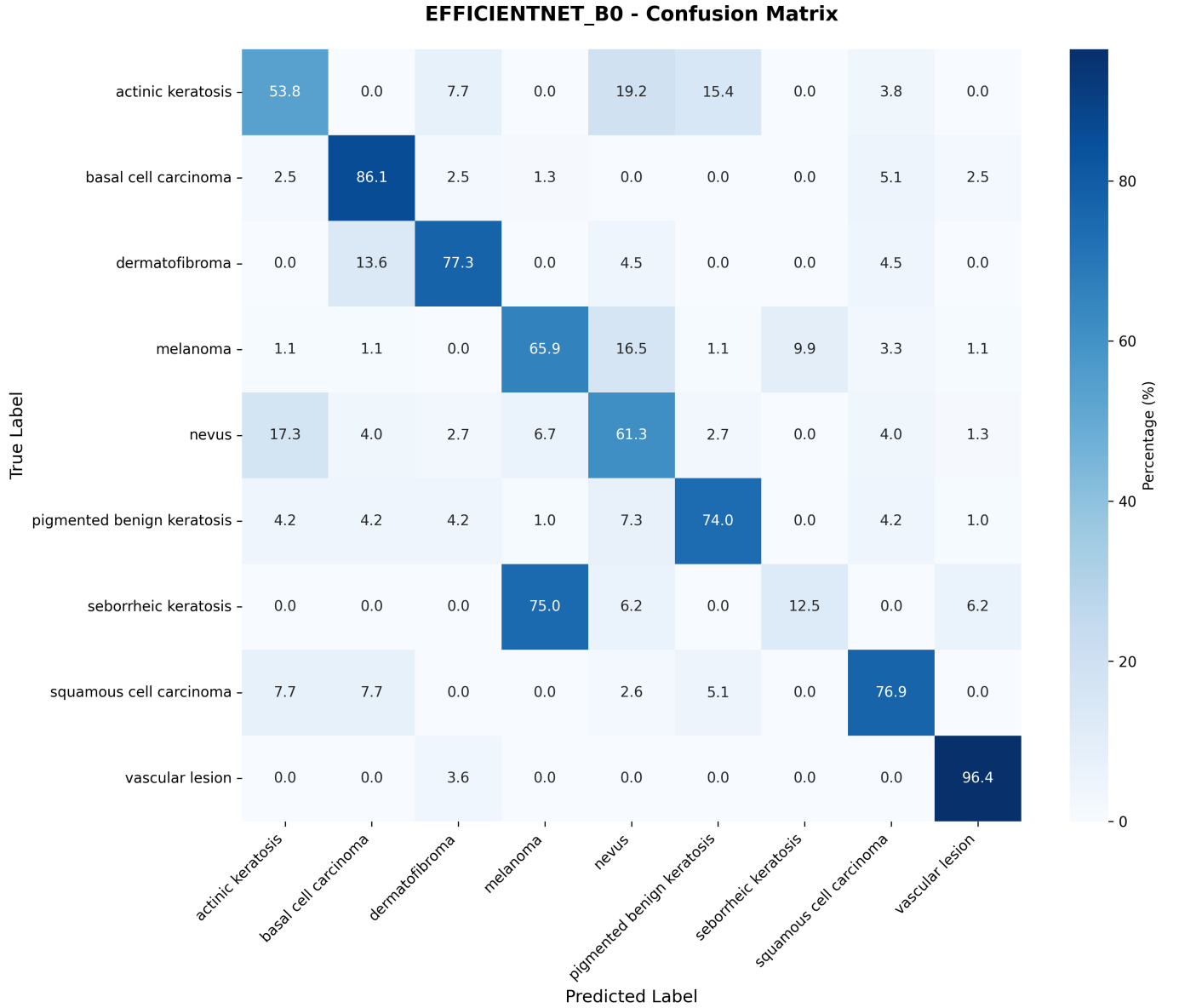


Fig. 3: EfficientNet-B0 confusion matrix showing superior balanced performance across all nine classes.

results highlights why external validation is critical in medical AI—models must perform well on completely unseen data, not just validation sets.

B. Model Performance and Generalization

EfficientNet-B0 was the best-performing architecture, achieving 70.97% test accuracy with excellent generalization. The model’s small validation-test gap (0.38%) shows it learned robust features without overfitting, likely because its parameter-efficient design (4.0M parameters) was well-suited for the limited training data (1,885 samples).

In contrast, ViT-B/16’s large size (85.8M parameters) led to substantial overfitting (4.99% gap). The model memorized validation patterns but could not generalize to new test data. This aligns with recent research showing that transformers

need much larger datasets than CNNs to reach their full potential [8]. The $21.5\times$ parameter difference between EfficientNet-B0 and ViT-B/16 (4.0M vs. 85.8M) shows that *parameter efficiency*, not raw size, determines performance on small datasets.

ResNet18 had lower overall accuracy but showed the most stable generalization (0.21% gap), making it a good choice when reliability matters more than peak performance. The model’s consistent cross-validation results (mean $64.68\% \pm 2.89\%$) and fast convergence (12 epochs) suggest it learned simpler features that transfer well but cannot make the fine-grained distinctions that EfficientNet-B0’s more complex architecture can.

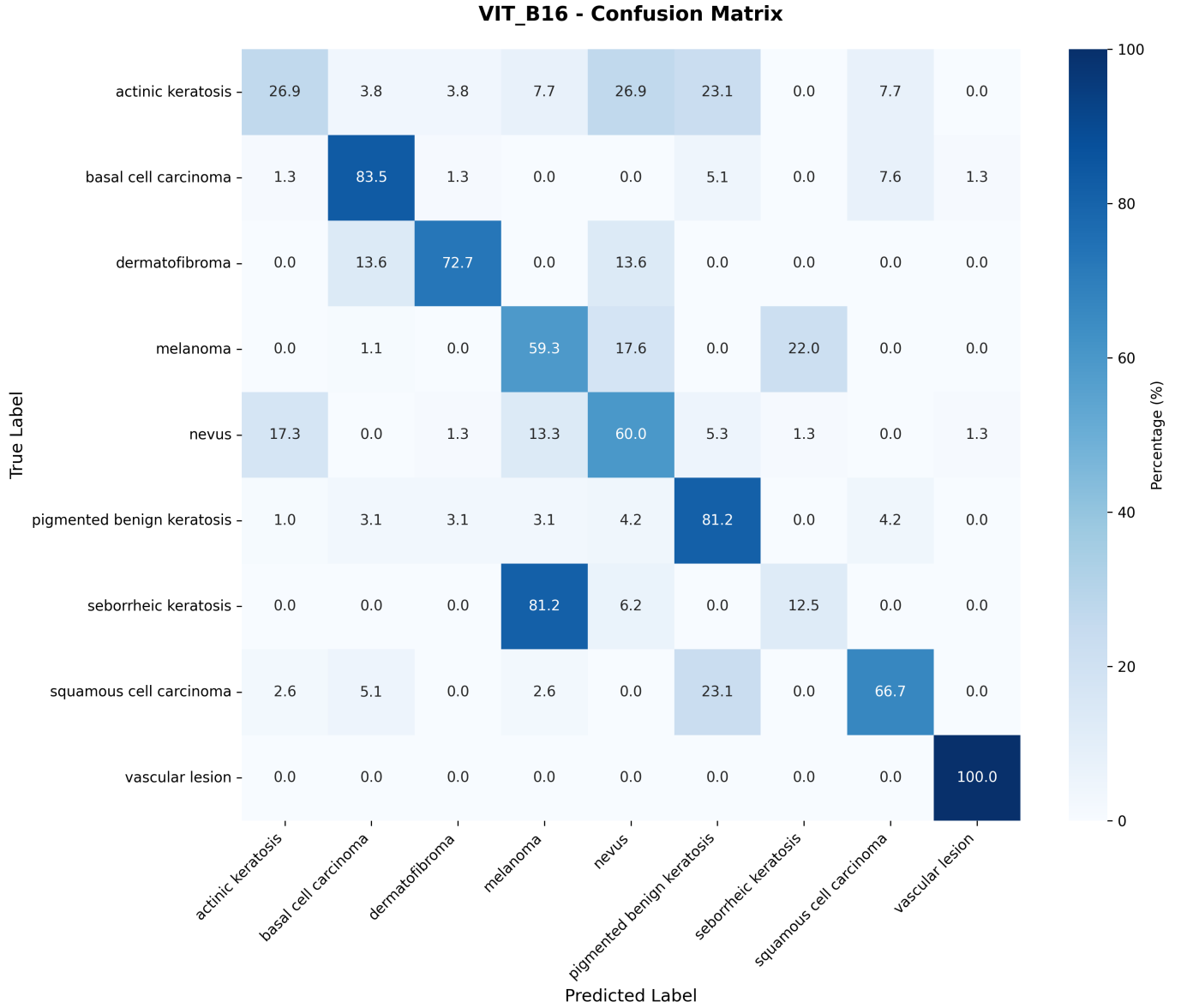


Fig. 4: ViT-B/16 confusion matrix revealing overfitting patterns with inconsistent per-class performance.

C. Clinical Implications

From a clinical perspective, EfficientNet-B0’s 81.28% malignant sensitivity represents a significant advantage. The model correctly identified 191 out of 235 malignant cases, detecting 21 additional cancers compared to ViT-B/16 (44 vs. 65 false negatives)—a 47% reduction in missed diagnoses.

To quantify this improvement’s potential clinical impact, consider a screening scenario with 10,000 patients where 50% have malignant lesions (5,000 malignant cases). The sensitivity difference between EfficientNet-B0 and ViT-B/16 is:

$$\Delta \text{Sensitivity} = 81.28\% - 72.34\% = 8.94\% \quad (2)$$

Applying this improvement rate to the population:

$$\text{Additional Detections} = 5,000 \times 0.0894 = 447 \text{ cases} \quad (3)$$

This translates to approximately **447 additional malignant lesions detected** in a 10,000-patient cohort, enabling earlier intervention and potentially better treatment outcomes. While this does not directly mean 447 lives saved—since survival depends on lesion stage, treatment effectiveness, and patient factors—earlier detection strongly correlates with better prognosis in melanoma and other skin cancers [1].

While ViT-B/16 had slightly higher specificity (78.90%), this came at the cost of unacceptably high false negatives for a cancer screening tool. Missing malignant lesions has far more serious consequences than false positives. Since early detection is crucial for treatment success and survival, sensitivity is the priority metric for clinical use.

The confusion matrices show that all models struggled with minority classes (*Seborrheic Keratosis*, *Dermatofibroma*),

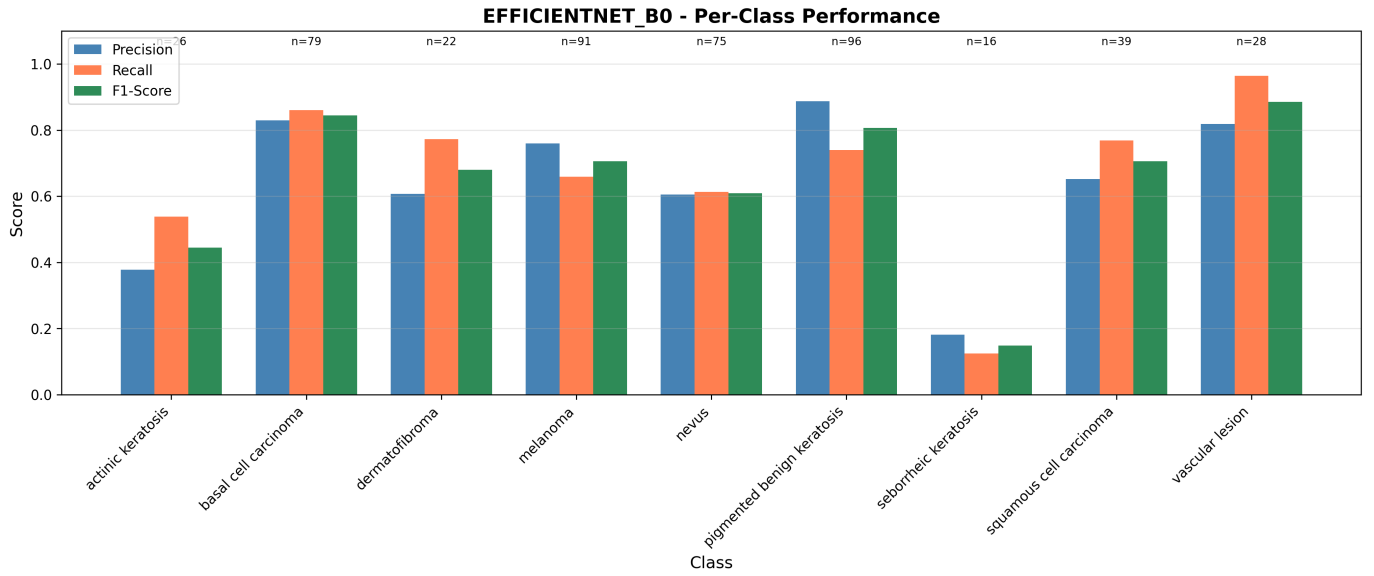


Fig. 5: Per-class performance metrics for EfficientNet-B0 showing precision, recall, and F1-score with sample counts. Minority classes (Seborrheic Keratosis, Dermatofibroma) exhibit lowest performance.

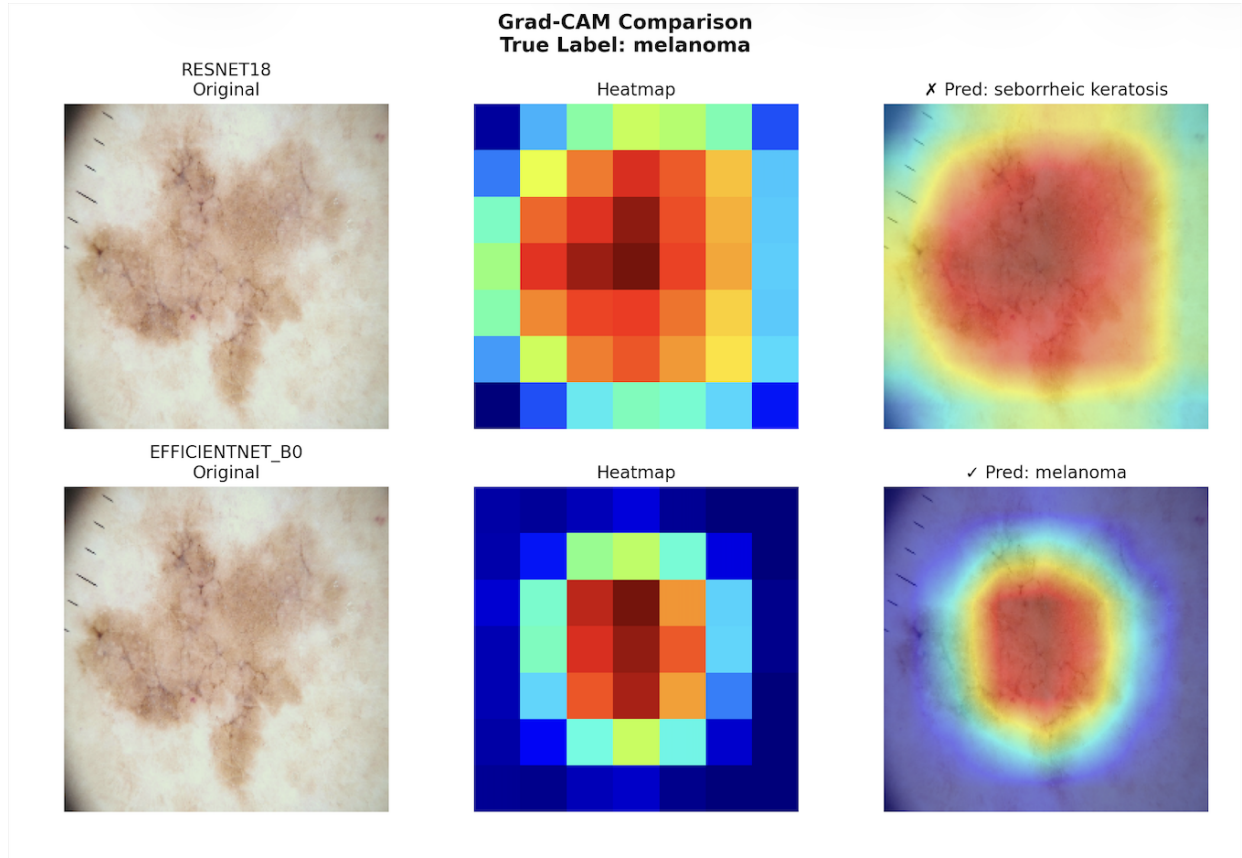


Fig. 6: Cross-model Grad-CAM comparison on a melanoma sample. ResNet18 (top) misclassified as seborrheic keratosis, while EfficientNet-B0 (bottom) correctly predicted melanoma with more focused attention on lesion boundaries.

highlighting the ongoing challenge of class imbalance despite using weighted loss functions. We found a *critical mass threshold*: classes with fewer than 100 training samples had below 50% precision across all models, showing that transfer

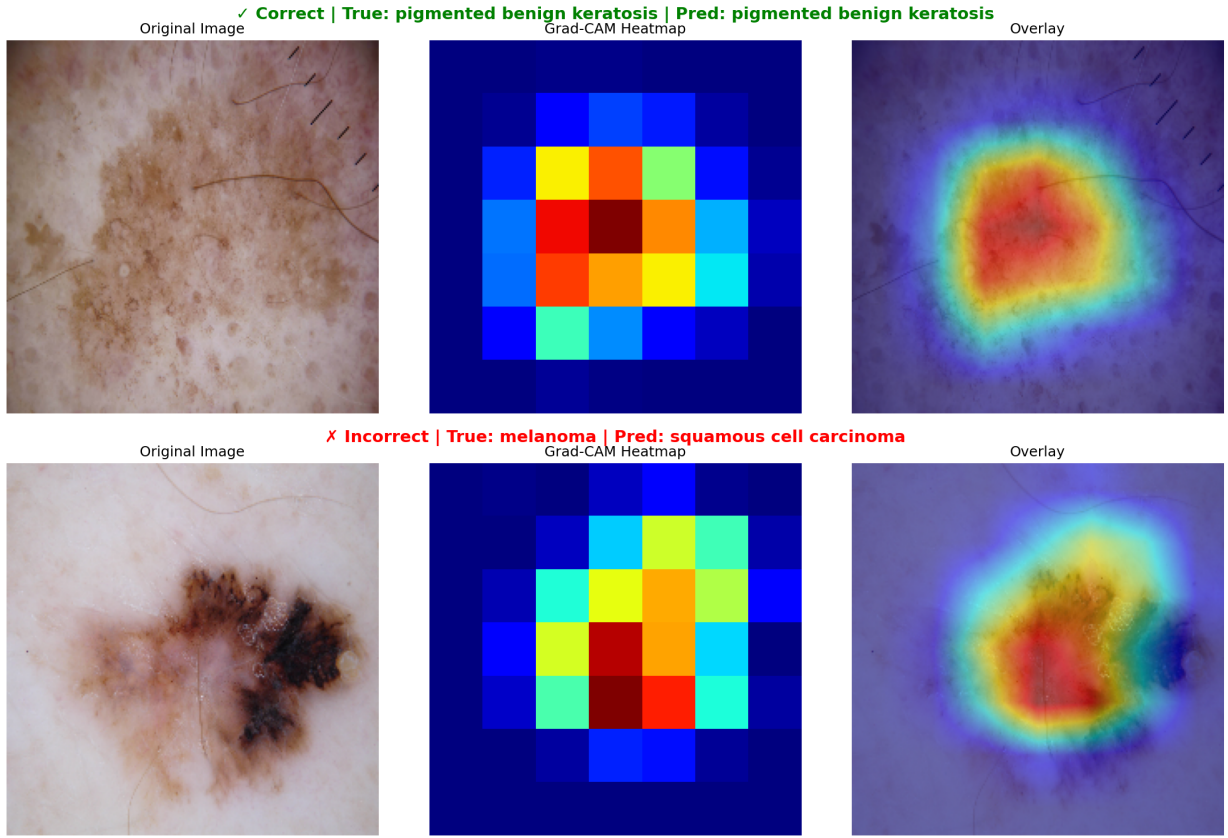


Fig. 7: Grad-CAM visualizations for EfficientNet-B0 showing attention regions for correct and incorrect predictions. Top: Correct prediction of Melanoma with focused attention on irregular borders and pigmentation. Bottom: Incorrect prediction (Nevus → Melanoma) showing diffuse attention pattern.

learning and data augmentation cannot fully compensate for very limited data. Future work could explore focal loss [10] or class-specific augmentation strategies to improve performance on minority classes.

D. Interpretability and Model Trust

Grad-CAM visualizations confirmed that EfficientNet-B0 learned clinically relevant features, focusing on lesion boundaries, asymmetry, and color variations—key diagnostic criteria from the dermatological ABCDE rule (Asymmetry, Border, Color, Diameter, Evolving). This interpretability is important for clinical adoption, since physicians need explanations for model predictions to build trust and validate decisions.

The match between the model’s attention patterns and established diagnostic criteria suggests that EfficientNet-B0 captures real visual features rather than spurious correlations. This allows clinicians to evaluate the model’s reliability on each case individually, seeing when the model’s reasoning matches or differs from their clinical judgment.

We could not generate meaningful Grad-CAM visualizations for ViT-B/16 because of its non-convolutional architecture, which is a practical limitation for medical applications where interpretability is essential. While attention rollout

methods exist for transformers, they are less intuitive than CNN-based activation maps and show less precise spatial information, making it harder for clinicians to validate model decisions based on specific anatomical features.

E. Limitations

Several limitations should be acknowledged:

1) Dataset Limitations:

- **Size:** 2,357 images insufficient for large architectures, as evidenced by ViT-B/16 overfitting.
- **Diversity:** Single-source data (ISIC) may not represent global patient demographics, skin types, or imaging equipment variations.
- **Class Imbalance:** Despite mitigation strategies, minority classes remain severely under-represented (less than 100 samples).

2) Methodological Limitations:

- **Single Dataset:** No external validation on independent datasets (e.g., HAM10000, Derm7pt).
- **Architecture Scope:** Only three models evaluated; modern architectures (e.g., ConvNeXt, Swin Transformer) not explored. Trying other models might have yielded more successful results.

3) Deployment Considerations:

- **Real-World Validation:** No prospective clinical trial; performance in actual clinical practice unknown.
- **Demographic Fairness:** No analysis of performance across different Fitzpatrick skin types—models may exhibit bias toward lighter skin tones overrepresented in training data.
- **Geographic Generalization:** Single-source dataset may not generalize to different populations, imaging protocols, or clinical settings.

F. Future Directions

While our results show that the model works well, several important areas need further investigation to improve clinical utility and robustness:

- **Ensemble Methods:** Combining EfficientNet-B0 with other models to improve overall robustness and generalization.
- **Better Interpretability:** Developing attention-based mechanisms designed specifically for medical imaging to keep models interpretable for clinicians.
- **Addressing Class Imbalance:** Using advanced techniques like focal loss [10] or class-specific data augmentation to handle imbalanced datasets better.
- **External Validation:** Testing the model on independent datasets from multiple medical centers (e.g., HAM10000, Derm7pt) to assess readiness for real-world use.
- **Adding Patient Information:** Incorporating patient metadata like age, lesion location, and medical history to improve diagnostic accuracy.

Beyond these technical improvements, ethical deployment is critical. We need comprehensive fairness analysis across demographic groups and Fitzpatrick skin types to ensure the model performs equally well for all patient populations. Finally, prospective clinical trials comparing model-assisted diagnosis with standard dermatological practice will provide the evidence needed for regulatory approval and clinical adoption.

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